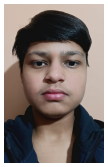
Aashirvad Bajpai



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OBJECTIVE

B.Tech IT student with experience in AI systems, GPU-accelerated pipelines, and open-source scientific computing. Seeking software engineering and AI-focused roles to build scalable, high-performance systems.

EDUCATION

Manipal Institute of Technology (MIT), MAHE Expected 2027
B.Tech in Information Technology CGPA: 8.04
Relevant Coursework: Data Structures, Algorithms, AI, ML

SKILLS

Languages	Python, C++, Java, SQL, C
AI/ML	PyTorch, TensorFlow, Scikit-learn, NumPy, Pandas, LangChain
Data Systems	Apache Arrow (PyArrow), Awkward Array, cuDF
Development	React.js, Node.js, MongoDB
Tools	Git, Linux, VS Code, Jupyter, LaTeX
Other	Competitive Programming, OOP Design,embedded systems

PROJECTS

Brain-to-Text Decoding (GPU Pipeline) – Developed a GPU-accelerated neural signal preprocessing and CTC-based sequence-to-text inference pipeline on NVIDIA A100. Implemented custom beam search decoding with KenLM-based language model rescoring to improve prediction coherence. Reduced validation loss from 1.97 to 0.89 across 20 epochs (8k+ trials) and optimized device–host transfers to sustain ∼1.3 it/s under GPU-bound workloads. GitHub

Knowledge Graph–to–Text Generation Research – Coauthor of IEEE MoSICom 2025 paper on ontology-driven triple-to-text generation using LLaMA-based models; extended manuscript under review at IEEE Open Journal of the Computer Society (OJCS).

Explainable Clinical Decision Support System for ICU Treatment Recommendation – Developed an interpretable, similarity-based CDSS on MIMIC-IV ICU data to identify comparable patients and recommend treatments from historical outcomes. Engineered leakage-safe patient representations from early ICU signals (vitals, labs, comorbidities) and applied cosine/Euclidean similarity with embedding extensions. Designed a transparent case-based pipeline with feature-level contribution analysis for auditability, clinical alignment, and safety in real-world settings.

CERTIFICATIONS

Machine Learning Foundations Coursera
Oracle Cloud Infrastructure 2025 Certified AI Foundations Associate Oracle
CITI Program – Data or Specimens Only Research (Human Subjects Research Ethics, HIPAA, IRB)
Certificate Verification

ACHIEVEMENTS & EXTRACURRICULARS

- College Finalist, **Smart India Hackathon 2024, 2025**
- Member, IEEE SBM; AI Researcher, MRM
- **Open-Source Contributions (Scikit-HEP, CERN-HSF, AIDASoft, OmniFold)** – Contributed to Awkward Array (CERN-HSF) with fixes in ArrayBuilder formatting, axis semantics docs, and GPU/cuDF compatibility; developing backend-agnostic byteswap (NumPy/CuPy/JAX), PyArrow string kernels, and cuDF $\geq$ 24.12 fixes. Diagnosed float64 precision issue in ak.to_json traced to C++ strtod, acknowledged by maintainers. Contributed to PODIO (AIDASoft) with JSON tests, FrameData backend prototype, and CI root-cause analysis influencing upstream fixes. Improved OmniFold compatibility with modern TensorFlow/Keras and validated execution stability.